

KLAES Legal Search Timeline: Developer Implementation Flow

1. Data Model & Entity Extraction

Before sorting, the backend must aggregate all events linked to the primary `TrackingID` / `FileNo`. The data fetch must include:

- Primary File Events: Commissioning, Instruments, Updates.
- Linked/Child Files: For Subdivisions, Mergers, Extensions.
- Parent Files: If the current file was born from a Decommissioned parent.
- KANGIS Linked Files: Files with prefixes `KNML`, `MLKN`, `KNGP`, `KN`.
- DCIV Linked Files: Investigation files linked to the primary file.
- Encumbrances: Caveats linked to specific instrument IDs.

2. Event Classification & Weight Assignment

Define an enumeration or constant map for the Timeline Weights. Every transaction fetched must be mapped to a weight.

```
```javascript
```

```
const TIMELINE_WEIGHTS = {
 FILE_COMMISSIONING: 12,
 TEMP_FILE_COMMISSIONING: 12,
 OCCUPANCY_PERMIT: 11,
 TRANSFER_OF_TITLE_OP: 10,
 RIGHT_OF_OCCUPANCY: 9,
 KANGIS_RECERTIFICATION: 8, // Implicit weight to ensure it
 sits above Cofo (0)
 CERTIFICATE_OF_OCCUPANCY: 0,
 OTHER_INSTRUMENTS: 0, // Assignments, Mortgages, etc.
 PARCEL_UPDATE: null, // Variable/Chronological
 TITLE_STATUS_UPDATE: null, // Variable/Chronological
 FILE_DECOMMISSIONING: null, // Variable/Chronological
```

```
 DCIV_COMMISSIONING: null // Variable/Chronological
};
...
```

### 3. The Sorting Algorithm (The "Timeline Weighting Method")

The timeline is not purely chronological. It uses a Composite Sort Key.

Sort Logic:

1. Primary Sort: `Weight` (Descending: `12` down to `0`).
2. Secondary Sort: `Event Timestamp` (Ascending: Oldest to Newest).
3. Tertiary Sort: `Positional Override Flags` (Boolean/Integer for special rules).

Note: Events with `null` weights (Parcel Updates, Title Status Updates, Decommissioning) will be treated as "Floating Events". They will be sorted purely chronologically but injected into the timeline without disrupting the weighted hierarchy of the main instruments.

### 4. Special Positional Rules & Overrides (Pre-Sort Processing)

Before applying the final sort, the backend must apply these business rules to adjust the `**sort\_index**` or `**parent\_id**` of specific events.

Rule A: Temporary File Nesting

- Condition: If a `Temporary File Commissioning` exists for the main file.
- Action: Force its `**sort\_index**` to be exactly `Main File Commissioning Index + 1`. It must render directly underneath the Main File Commissioning line.

Rule B: KANGIS Recertification & CofO Ingestion

- Condition: If the file is a KANGIS file (or ingested by one) and has both a `KANGIS Recertification` event and a `KANGIS CofO`.
- Action: The `KANGIS Recertification` timestamp/weight must be strictly forced to precede the `KANGIS CofO`.

- Logic: If `KANGIS\_Recert.date` >= `KANGIS\_CofO.date`, adjust `KANGIS\_Recert.date` to `KANGIS\_CofO.date - 1 second` for sorting purposes only.

Rule C: File Decommissioning & Commissioning Pairs (Parcel Updates)

- Condition: A Parcel Update (e.g., Subdivision) triggers the Decommissioning of File A and the Commissioning of File B.
- Action:
  1. File A's original Commissioning stays at the top (Weight 12).
  2. The Parcel Update event, File A's Decommissioning, and File B's Commissioning share the same chronological timestamp.
  3. Render Order at that timestamp: Parcel Update Event  $\rightarrow$  File A Decommissioning  $\rightarrow$  File B Commissioning (Weight 12).

Rule D: Encumbrances (Caveats)

- Condition: A Caveat is placed on the file.
- Action: DO NOT create a standalone timeline row for the Caveat.
- Logic: Query the `Caveat` table, find the `Instrument\_ID` it was placed against, and append the Caveat details to the `comments` array of that specific Instrument's timeline object.

## 5. JSON Output Structure (API Response)

The backend should return a structured JSON array that the frontend can render directly.

```
```json
{
  "fileNo": "RES-2020-1234",
  "trackingID": "TRK-998877",
  "timeline": [
    {
```

```
"id": 1,
"eventType": "FILE_COMMISSIONING",
"eventName": "File Commissioning",
"date": "2020-01-15T09:00:00Z",
"weight": 12,
"details": "Commissioned by Land Department",
"comments": []
},
{
  "id": 2,
  "eventType": "TEMP_FILE_COMMISSIONING",
  "eventName": "Temporary File Commissioning",
  "date": "2020-01-16T10:00:00Z",
  "weight": 12,
  "details": "Temp File No: TEMP-5544",
  "comments": []
},
{
  "id": 3,
  "eventType": "OCCUPANCY_PERMIT",
  "eventName": "Occupancy Permit (OP)",
  "date": "2020-05-20T14:00:00Z",
  "weight": 11,
  "details": "Issued to John Doe",
  "comments": []
},
{
```

```
"id": 4,
"eventType": "TRANSFER_OF_TITLE_OP",
"eventName": "Transfer of Title (OP)",
"date": "2021-03-10T11:00:00Z",
"weight": 10,
"details": "Transferred to Jane Smith",
"comments": []
},
{
  "id": 5,
  "eventType": "PARCEL_UPDATE",
  "eventName": "Subdivision",
  "date": "2022-06-15T09:00:00Z",
  "weight": null,
  "details": "Plot subdivided into 2 units",
  "comments": []
},
{
  "id": 6,
  "eventType": "FILE_DECOMMISSIONING",
  "eventName": "File Decommissioning (Parent)",
  "date": "2022-06-15T09:05:00Z",
  "weight": null,
  "details": "Decommissioned due to Subdivision",
  "comments": []
},
{
```

```

    "id": 7,
    "eventType": "RIGHT_OF_OCCUPANCY",
    "eventName": "Right of Occupancy (RofO)",
    "date": "2023-01-20T10:00:00Z",
    "weight": 9,
    "details": "RofO Issued",
    "comments": [
      {
        "type": "CAVEAT",
        "text": "Caveat placed by First Bank PLC on 2023-02-
01"
      }
    ]
  },
  {
    "id": 8,
    "eventType": "KANGIS_RECERTIFICATION",
    "eventName": "KANGIS Recertification",
    "date": "2024-08-10T08:00:00Z",
    "weight": 8,
    "details": "File ingested by KANGIS (KNML-2024-001)",
    "comments": []
  },
  {
    "id": 9,
    "eventType": "CERTIFICATE_OF_OCCUPANCY",
    "eventName": "Certificate of Occupancy (CofO)",
    "date": "2024-08-10T08:05:00Z",

```

```

    "weight": 0,
    "details": "KANGIS Cofo Issued",
    "comments": []
  },
  {
    "id": 10,
    "eventType": "TITLE_STATUS_UPDATE",
    "eventName": "Amendment (Title)",
    "date": "2025-02-14T12:00:00Z",
    "weight": null,
    "details": "Name corrected on Title",
    "comments": []
  }
]
}
...

```

6. Frontend Rendering Guidelines

1. Vertical Timeline UI: Render the array sequentially from top to bottom.

2. Weight Visuals:

Weight `12` to `9`: Use prominent, bold icons (e.g., Gold/Green milestones).

Weight `8`: Use a distinct KANGIS-branded icon.

Weight `0` & `null`: Use standard, lighter timeline dots.

3. Caveat Display: If the `comments` array contains a `CAVEAT` object, render it as a nested, highlighted warning block inside the instrument's timeline card, not as a separate line.

4. Linked Files: If a `FILE\_DECOMMISSIONING` or `PARCEL\_UPDATE` references a new `FileNo`, make that `FileNo` a clickable hyperlink that opens the Legal Search for the new file.

Temporary Files should automatically display the File Commissioning of the Main File and ALL the transaction of the Main File likewise the Main File should also display the commissioning of the Temporary File and also ALL the transactions referencing the Temporary File.